

Q1 - Output

DS Lab 05 - Stack and Queues

Q1 - Output

```
Enter number of elements: 6
Enter 6 elements in bottom-to-top order:
7 2 9 4 1 6
```

```
Stack (bottom -> top): 7 2 9 4 1 6
Maximum value: 9
Minimum value: 1
Middle elements: 9 and 4
```

Q2 - Output

DS Lab 05 - Stack and Queues

Q2 - Output

```
Enter size of Stack 1: 4
Enter size of Stack 2: 6
Enter Stack 1 elements in bottom-to-top order:
1 2 3 4
Enter Stack 2 elements in bottom-to-top order:
6 8 7 5 9 10

Third stack (bottom -> top): 1 6 2 8 3 7 4 5 9 10
Sorted elements (ascending): 1 2 3 4 5 6 7 8 9 10
Circular queue (front -> rear): 1 2 3 4 5 6 7 8 9 10
```

Q3 Bonus - Output

1. Double-Angle Formula

RHS Infix : $\pm(\sqrt{(1-\cos(\theta))/(1+\cos(\theta))})$

Postfix : 1 theta cos - 1 theta cos + / sqrt +/-

Prefix : +/- sqrt / - 1 cos theta + 1 cos theta

2. Pythagorean Identity

RHS Infix : $1-\cos(\theta)^2$

Postfix : 1 theta cos 2 ^ -

Prefix : - 1 ^ cos theta 2

3. Product-to-Sum Identity

RHS Infix : $(1/2)(\cos(\alpha-\beta)-\cos(\alpha+\beta))$

Postfix : 1 2 / alpha beta - cos alpha beta + cos - *

Prefix : * / 1 2 - cos - alpha beta cos + alpha beta

Q4 - Output

Q4 - Output

```
Enter number of pieces: 2
Enter Expression 1: x^2 + 2*x + 1
Enter Condition 1: x >= 0
Enter Expression 2: 2*x^2 - x + 3
Enter Condition 2: x < 0
Enter x: 5

Selected condition: x >= 0
Selected expression: x^2 + 2*x + 1
Postfix expression: x 2 ^ 2 x * + 1 +
Calculated value: 36.00
```

Q5 - Output

DS Lab 05 - Stack and Queues



Q5 - Output

```
Enter number of terms n: 5
Series: 1 + 4 + 7 + 10 + 13
Postfix: 1 4 + 7 + 10 + 13 +
Prefix: + + + + 1 4 7 10 13
```